



**Tertiary Infotech Academy Pte Ltd**

UEN: 201200696W

# LESSON PLAN

For

## **Fast-Track Innovations with Agile Design Thinking and Generative AI (GenAI)**

TGS Ref No: TGS-2024049781

Conducted by

**Tertiary Infotech Academy Pte Ltd**

UEN: 201200696W

**Version 4.0**

## DOCUMENT VERSION CONTROL RECORD

| Version Number | Effective Date of Release | Summary of Included Changes                                                                                                                                                                                                                                                                                                                                                                                                                          | Author         |
|----------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 3.0            | 1 January 2026            | Legacy master trainer deck and lesson plan (v3) — 4 topics, exercise-based delivery.                                                                                                                                                                                                                                                                                                                                                                 | Dr. Alfred Ang |
| 4.0            | 17 August 2026            | Major revision. Content beefed up from Thoughtworks, Adaptovate, BMC, PMI, UX Magazine, The Design Gym, PremierAgile and Startup Frontier. Twelve real-world case-study activities (Airbnb, Netflix/Blockbuster, GE Adventure Series, IDEO, Spotify, IBM, DBS, Boeing 737 MAX) replace the generic exercises, each with scenario, discussion questions and debrief. Design Thinking Studio and Padlet Classroom Board adopted as the class ed-tools. | Dr. Alfred Ang |

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## Course Information

|                             |                                                                             |
|-----------------------------|-----------------------------------------------------------------------------|
| <b>Course Title</b>         | Fast-Track Innovations with Agile Design Thinking and Generative AI (GenAI) |
| <b>WSQ Course Reference</b> | TGS-2024049781                                                              |
| <b>Training Provider</b>    | Tertiary Infotech Academy Pte Ltd (UEN 201200696W)                          |
| <b>Duration</b>             | 2 days · 8 training hours per day (16 hours) + 2 hours assessment           |
| <b>Daily Timing</b>         | 9:30 am – 6:30 pm (1-hour lunch; tea breaks counted within training time)   |
| <b>Mode</b>                 | Instructor-led, case-study based, with collaborative online ed-tools        |
| <b>Skills Framework</b>     | Design Thinking Practice (ICT-ACE-5014-1.1)                                 |
| <b>Trainer</b>              | Dr. Alfred Ang                                                              |

## Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Integrate design thinking methodologies and agile principles to drive organisational innovation using generative AI.
- LO2: Synthesise stakeholder inputs to uncover end-user needs for successful innovation.
- LO3: Lead design thinking projects using project management tools and techniques to enhance organisational performance.
- LO4: Develop strategies for agile design thinking to enhance product and service innovation.

## Skills Framework Alignment

TSC Title: Design Thinking Practice · TSC Code: ICT-ACE-5014-1.1

Abilities:

- A1 – Integrate design thinking methodologies into processes to drive innovation across the organisation
- A2 – Develop strategies to proliferate design thinking across the organisation
- A3 – Synthesise information from different sources and stakeholders in order to fully understand the needs of end users
- A4 – Drive the development of new strategies to enhance products and/or services for the organisation
- A5 – Engage stakeholders during the design thinking process to uncover the motivations behind their actions and behaviours
- A6 – Cultivate design thinking as a viable tool and methodology to foster new innovations for the organisation
- A7 – Lead design thinking projects across the organisation

Knowledge:

- K1 – Latest trends in design thinking
- K2 – Concept of innovation management
- K3 – Drivers of organisational growth and success
- K4 – Concept and principles of resource management
- K5 – Project management tools and techniques

## Teaching and Learning Methods

- Trainer-led concept delivery using a highly visual slide deck (diagrams, comparison tables and frameworks rather than bullet walls).
- Twelve real-world case-study activities, each carrying a documented case, a workplace scenario, five discussion questions and a trainer debrief.
- Collaborative group work in groups of 3–5 using two online ed-tools: the Design Thinking Studio (5-stage workspace) and the Padlet Classroom Board.
- Guided use of generative AI tools (ChatGPT / Microsoft Copilot) with explicit critique of AI output against evidence.
- Cross-group peer review — groups critique each other's output and defend their reasoning.
- Continuous formative feedback from the trainer during each activity, and a structured debrief after each one.

## Learning Resources

- Trainer Slides (PPT/PDF), Learner Guide (PDF) and this Lesson Plan (PDF) — downloadable from the LMS at <https://lms-tms.tertiaryinfotech.com>.
- Twelve individual activity briefs (PDF), one per activity, each containing the case, the scenario, the discussion questions, the step-by-step instructions and the debrief.
- Design Thinking Studio — <https://alfredang.github.io/designthinking/>
- Padlet Classroom Board — <https://alfredang.github.io/padlet/>
- A generative AI assistant (ChatGPT or Microsoft Copilot) and a laptop with a modern web browser.

## Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Case Study (CS) — a scenario-based innovation case, 1 hour, open book.
- Format: Open Book — course slides, Learner Guide, activity briefs and approved materials only.
- The final assessment is conducted at the end of Day 2.
- Learners are assessed as Competent (C) or Not Yet Competent (NYC) on each instrument.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

## Course Schedule

### Day 1 — Foundations, Problem Framing and Ideation with GenAI

| Time        | Duration | Topic / Activity                                                                                                                                                                                                                                    |
|-------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9:30–10:00  | 30 min   | Welcome, trainer and learner introductions, ground rules, learning outcomes, course outline and mandatory SSG digital attendance (AM). Introduction to the two course ed-tools: Design Thinking Studio and the Padlet Classroom Board.              |
| 10:00–11:00 | 60 min   | <b>Topic 1 — Foundations of Design Thinking, Agile and Generative AI: what design thinking is; the three mindsets (Design Thinking, Lean, Agile) and the question each answers; the Double Diamond; the five stages; wicked problems.</b>           |
| 11:00–11:15 | 15 min   | Tea break                                                                                                                                                                                                                                           |
| 11:15–12:00 | 45 min   | <b>Topic 1 continued — Product development through design thinking and agile; how generative AI enhances the loop; where AI must not replace human judgement; latest trends in innovation.</b>                                                      |
| 12:00–13:00 | 60 min   | Hands-on case study: Activity 1: Airbnb — Diagnosing Why a Method Alone Does Not Save a Business                                                                                                                                                    |
| 13:00–14:00 | 60 min   | Lunch break                                                                                                                                                                                                                                         |
| 14:00–14:15 | 15 min   | Mandatory SSG digital attendance (PM)                                                                                                                                                                                                               |
| 14:15–15:30 | 75 min   | Hands-on case studies: Activity 2: Netflix vs Blockbuster — Innovation as an Operating System; Activity 3: Positioning GenAI in the Innovation Loop — Where It Helps and Where It Lies                                                              |
| 15:30–15:45 | 15 min   | Tea break                                                                                                                                                                                                                                           |
| 15:45–16:45 | 60 min   | <b>Topic 2 — Problem Framing and Ideation: framing the design challenge; the Problem-Assumption Model; problem statement vs Point of View vs How Might We; the altitude test; empathy maps and persona mapping with GenAI; ideation techniques.</b> |
| 16:45–18:15 | 90 min   | Hands-on case studies: Activity 4: GE Adventure Series —                                                                                                                                                                                            |

|             |        |                                                                                                              |
|-------------|--------|--------------------------------------------------------------------------------------------------------------|
|             |        | Reframing the Problem Instead of Fixing the Machine; Activity 5: IDEO Shopping Cart — Volume, Then Judgement |
| 18:15–18:30 | 15 min | Day 1 recap, Q&A and close                                                                                   |

Total training time: 480 minutes (8 hours).

## Day 2 — Agile Delivery, Scaling Innovation and Assessment

| Time        | Duration | Topic / Activity                                                                                                                                                                                                                                                                                                           |
|-------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9:30–9:45   | 15 min   | Day 1 recap and mandatory SSG digital attendance (AM)                                                                                                                                                                                                                                                                      |
| 9:45–10:45  | 60 min   | Topic 2 concluded — prototyping with AI-driven tools and defining a falsifiable test plan. Hands-on case study: Activity 6: Rapid Prototyping with AI — From Concept to Testable Artefact in One Hour                                                                                                                      |
| 10:45–11:00 | 15 min   | Tea break                                                                                                                                                                                                                                                                                                                  |
| 11:00–12:00 | 60 min   | <b>Topic 3 — Agile Development and AI for Rapid Solution Delivery: being agile vs doing agile; the Scrum loop and the three accountabilities; dual-track agile; the mini-waterfall anti-pattern; Epic → Feature → Hill → Story → Task.</b>                                                                                 |
| 12:00–13:00 | 60 min   | Hands-on case study: Activity 7: Spotify — Reading a Scaling Model Honestly                                                                                                                                                                                                                                                |
| 13:00–14:00 | 60 min   | Lunch break                                                                                                                                                                                                                                                                                                                |
| 14:00–14:15 | 15 min   | Mandatory SSG digital attendance (PM)                                                                                                                                                                                                                                                                                      |
| 14:15–15:30 | 75 min   | Topic 3 continued — user stories and the 3 C's, the three-cake release model, story points and Planning Poker, GenAI across the sprint. Hands-on case studies: Activity 8: From Hills to Backlog — Converting Ideas into Stories with GenAI; Activity 9: Running Dual-Track Agile — A Sprint Simulation with AI Assistance |
| 15:30–15:45 | 15 min   | Tea break                                                                                                                                                                                                                                                                                                                  |
| 15:45–16:45 | 60 min   | <b>Topic 4 — Scaling and Sustaining Innovations: the three levers that scale innovation; culture and the four principles; stakeholder buy-in; resource management with AI; sensemaking; systems thinking and feedback loops; metrics and KPIs.</b>                                                                         |
| 16:45–17:45 | 60 min   | Hands-on case studies: Activity                                                                                                                                                                                                                                                                                            |

|             |        |                                                                                                                                                                                                                    |
|-------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             |        | 10: DBS Bank — Scaling Innovation Across 25,000 People; Activity 11: Boeing 737 MAX — Systems Thinking and the Cost of Local Optimisation; Activity 12: Innovation Metrics — Designing KPIs That Change a Decision |
| 17:45–18:00 | 15 min | Course revision, Q&A, mandatory TRAQOM course feedback survey and Assessment digital attendance                                                                                                                    |
| 18:00–18:15 | 15 min | <b>Briefing for Assessment</b>                                                                                                                                                                                     |
| 18:15–18:30 | 15 min | <b>Final Assessment begins — Written Assessment (WA, SAQ) 1 hour then Case Study (CS) 1 hour, both open book. Learners submit on the LMS and sign the Assessment Summary Record.</b>                               |

Total training time: 480 minutes (8 hours).

## Activity Reference (aligned to topics and learning outcomes)

| Topic                                                                                            | Weighting | Activities                            | Ed-tool                         |
|--------------------------------------------------------------------------------------------------|-----------|---------------------------------------|---------------------------------|
| <b>Topic 01: Foundations of Design Thinking, Agile, and Generative AI for Problem-Solving</b>    | 25%       | Activity 1, Activity 2, Activity 3    | Padlet                          |
| <b>Topic 02: Problem Framing and Ideation: Leveraging Design Thinking and Generative AI</b>      | 25%       | Activity 4, Activity 5, Activity 6    | Design Thinking Studio          |
| <b>Topic 03: Agile Development and AI for Rapid Solution Delivery</b>                            | 25%       | Activity 7, Activity 8, Activity 9    | Design Thinking Studio / Padlet |
| <b>Topic 04: Scaling and Sustaining Innovations with Agile Design Thinking and Generative AI</b> | 25%       | Activity 10, Activity 11, Activity 12 | Design Thinking Studio / Padlet |

## Detailed Activity Schedule

| # | Activity                                                        | Duration | Real-world case                                  |
|---|-----------------------------------------------------------------|----------|--------------------------------------------------|
| 1 | Airbnb — Diagnosing Why a Method Alone Does Not Save a Business | 45 min   | Airbnb, 2009...                                  |
| 2 | Netflix vs Blockbuster                                          | 45 min   | In 2000 Reed Hastings offered to sell Netflix to |



|           |                                                                             |        |                                                                                               |
|-----------|-----------------------------------------------------------------------------|--------|-----------------------------------------------------------------------------------------------|
|           | — Innovation as an Operating System                                         |        | Blockbuster for \$50 million...                                                               |
| <b>3</b>  | Positioning GenAI in the Innovation Loop — Where It Helps and Where It Lies | 40 min | A Singapore bank's innovation team used a large language model to generate customer person... |
| <b>4</b>  | GE Adventure Series — Reframing the Problem Instead of Fixing the Machine   | 50 min | GE Healthcare's MRI scanners were technically excellent, but designer Doug Dietz watched a... |
| <b>5</b>  | IDEO Shopping Cart — Volume, Then Judgement                                 | 50 min | In 1999, ABC's Nightline gave the design firm IDEO five days to redesign the supermarket s... |
| <b>6</b>  | Rapid Prototyping with AI — From Concept to Testable Artefact in One Hour   | 50 min | A hospital innovation team in Singapore needed to test whether a self-service kiosk would ... |
| <b>7</b>  | Spotify — Reading a Scaling Model Honestly                                  | 45 min | In 2012 Henrik Kniberg and Anders Ivarsson published a paper describing how Spotify organi... |
| <b>8</b>  | From Hills to Backlog — Converting Ideas into Stories with GenAI            | 55 min | IBM runs a two-day Design Thinking and Agile workshop in which Day 1 produces 'Hills' — ou... |
| <b>9</b>  | Running Dual-Track Agile — A Sprint Simulation with AI Assistance           | 50 min | Marty Cagan describes the most common failure he sees in teams that believe they are agile... |
| <b>10</b> | DBS Bank — Scaling Innovation Across 25,000 People                          | 50 min | Around 2014 DBS Bank set out to 'Make Banking Joyful' and to become a technology company d... |
| <b>11</b> | Boeing 737 MAX — Systems Thinking and the Cost of Local Optimisation        | 50 min | Boeing needed a competitive response to the Airbus A320neo quickly and cheaply...             |
| <b>12</b> | Innovation Metrics — Designing KPIs That Change a Decision                  | 45 min | Two innovation programmes reported to the same board...                                       |